Picture Archiver and communication System A.K.A PACS

Introduction:

An imaging solution for Healthcare sector which helps the practitioners and radiologists to examine and review the studies (images from modalities). This solution acts as an image storage server which stores the images in formatted studies.

Pre requisites:
An Ubuntu 22.04 machine.

Internet for docker installation and configuration.

Steps:

1. sudo apt-get update
2. sudo apt-get upgrade

Install Docker

1- sudo apt install apt-transport-https ca-certificates curl software-properties-common
2- curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /usr/share/keyrings/docker-archive-keyring.gpg
3- echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/docker-archive-keyring.gpg] https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
4- sudo apt update
5- apt-cache policy docker-ce
6- sudo apt install docker-ce
7- sudo systemctl status docker
8- sudo usermod -aG docker ${USER}
9- su - ${USER}
10- groups

Now we will pull DCM4cHEE docker image and install with the docker-compose method. The site link is as below

https://github.com/dcm4che/dcm4chee-arc-light/

We are installing the non-secured minimal install
Add following groups (this is optional)

1. $ sudo -i
2. # groupadd -r slapd-dcm4chee --gid=1021 && useradd -r -g slapd-dcm4chee --uid=1021 slapd-dcm4chee
3. # groupadd -r postgres-dcm4chee --gid=999 && useradd -r -g postgres-dcm4chee --uid=999 postgres-dcm4chee
4. # groupadd -r dcm4chee-arc --gid=1023 && useradd -r -g dcm4chee-arc --uid=1023 dcm4chee-arc
5. # exit

Now create a directory dcm4chee in the opt directory

1- cd /opt
2- sudo mkdir –p /dcm4chee
3- sudo touch docker-compose.yml
4- sudo nano docker-compose.yml

Edit the docker-compose.yml file with following

```
version: "3"
services:
  ldap:
    image: dcm4che/slapd-dcm4chee:2.6.7-33.1
    logging:
      driver: json-file
      options:
        max-size: "10m"
    ports:
      - "389:389"
    environment:
```

```
      STORAGE_DIR: /storage/fs1
    volumes:
      - /var/local/dcm4chee-arc/ldap:/var/lib/openldap/openldap-data
      - /var/local/dcm4chee-arc/slapd.d:/etc/openldap/slapd.d
  db:
    image: dcm4che/postgres-dcm4chee:17.1-33
    logging:
      driver: json-file
      options:
        max-size: "10m"
    ports:
     - "5432:5432"
    environment:
      POSTGRES_DB: pacsdb
      POSTGRES_USER: pacs
      POSTGRES_PASSWORD: pacs
    volumes:
      - /etc/localtime:/etc/localtime:ro
      - /etc/timezone:/etc/timezone:ro
      - /var/local/dcm4chee-arc/db:/var/lib/postgresql/data
  arc:
    image: dcm4che/dcm4chee-arc-psql:5.33.1
    logging:
      driver: json-file
      options:
        max-size: "10m"
    ports:
      - "8080:8080"
      - "8443:8443"
      - "9990:9990"
      - "9993:9993"
      - "11112:11112"
      - "2762:2762"
      - "2575:2575"
      - "12575:12575"
    environment:
      POSTGRES_DB: pacsdb
```

```
    POSTGRES_USER: pacs
    POSTGRES_PASSWORD: pacs
    WILDFLY_CHOWN: /storage
    WILDFLY_WAIT_FOR: ldap:389 db:5432
  depends_on:
    - ldap
    - db
  volumes:
    - /etc/localtime:/etc/localtime:ro
    - /etc/timezone:/etc/timezone:ro
    - /var/local/dcm4chee-arc/wildfly:/opt/wildfly/standalone
    - /var/local/dcm4chee-arc/storage:/storage
```

Install docker-compose and pull dcm4chee image

1- sudo apt install docker-compose

   The pull the dcm4chee image using the following command

2- sudo docker-compose –p dcm4chee up –d

Make a host entry in the /etc/hosts file

1- sudo nano /etc/hosts
   ip and hostname

Install Portainer to handle containers easily. It is a web based tool to handle dockers.

1- sudo groupadd docker
2- sudo usermod -aG docker $USER
3- newgrp docker
4- docker ps
5-

```
sudo docker volume create portainer_data
sudo docker run -d -p 9000:9000 --name=portainer --
restart=always \
```

```
-v /var/run/docker.sock:/var/run/docker.sock \
-v portainer_data:/data \
portainer/portainer-ce
```

Access portainer at ip:9000

Now install OHIF-Viewer

1- Login to portainer and go to volumes
2- Create a volume and name it ohif and select local
3- Create container and name it ohif-viewer
4- Pull latest image or as you like I have used ohif/app:latest
5- Port mapping
   Host 3000 and container 80
6- Edit network section at the end and select dcm4chee_default network
7- Edit volumes tab with following location

- Container /usr/share/nginx/html
- Volume ohif-local

8- Now hit the deploy the container tab

Now edit the app-config.js file present in ohif-viewer container at /usr/share/nginx/html through portainer

The configurations can be found here
https://docs.ohif.org/configuration/dataSources/dicom-web

Select Dicomweb under Configuration tab, scroll through the page until you find this

```
window.config = {
  routerBasename: null,
```

```
  extensions: [],
  modes: [],
  showStudyList: true,
  dataSources: [
    {
      namespace: '@ohif/extension-default.dataSourcesModule.dicomweb',
      sourceName: 'dicomweb',
      configuration: {
        friendlyName: 'dcmjs DICOMWeb Server',
        name: 'DCM4CHEE',
        wadoUriRoot: 'https://server.dcmjs.org/dcm4chee-
arc/aets/DCM4CHEE/wado',
        qidoRoot: 'https://server.dcmjs.org/dcm4chee-arc/aets/DCM4CHEE/rs',
        wadoRoot: 'https://server.dcmjs.org/dcm4chee-arc/aets/DCM4CHEE/rs',
        qidoSupportsIncludeField: true,
        supportsReject: true,
        imageRendering: 'wadors',
        thumbnailRendering: 'wadors',
        enableStudyLazyLoad: true,
        supportsFuzzyMatching: true,
        supportsWildcard: true,
      },
    },
  ],
  defaultDataSourceName: 'dicomweb',
};
```

Edit this file by entering your ip and port

```
window.config = {
  routerBasename: null,
  extensions: [],
  modes: [],
  showStudyList: true,
  dataSources: [
    {
```

```
      namespace: '@ohif/extension-default.dataSourcesModule.dicomweb',
      sourceName: 'dicomweb',
      configuration: {
        friendlyName: 'dcmjs DICOMWeb Server',
        name: 'DCM4CHEE',
        wadoUriRoot: 'http://ip:8080/dcm4chee-arc/aets/DCM4CHEE/wado',
        qidoRoot: 'http:// ip:8080/dcm4chee-arc/aets/DCM4CHEE/rs',
        wadoRoot: 'http:// ip:8080/dcm4chee-arc/aets/DCM4CHEE/rs',
        qidoSupportsIncludeField: true,
        supportsReject: true,
        imageRendering: 'wadors',
        thumbnailRendering: 'wadors',
        enableStudyLazyLoad: true,
        supportsFuzzyMatching: true,
        supportsWildcard: true,
      },
    },
  ],
  defaultDataSourceName: 'dicomweb',
};
```

If you want to test your setup download dicom sample images from net and upload using dcm4chee server like
http://ip:8080/dcm4chee-arc/ui2
once in the server upload dcm4chee images, if the images are uploaded successfully you can view it using the ohif-viewer by following

http://Ip:3000
Here the server is up running, now if you want to integrate the modalities just supply the modalities with the PACS ip and port. Port is usually 11112 and service is DCM$CHEE.

That's all Folks !!!!